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Clockwise rotation of the Baoshan Block due to southeastward tectonic escape of Tibetan crust since the Oligocene

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SUMMARY

Understanding the mode of deformation around the Eastern Himalayan Syntaxis (EHS) is crucial for models of Tibetan Plateau evolution. To constrain rotations and translation east of the EHS, we present new palaeomagnetic data from meta-basalt layers—possibly sills—of the Baoshan Block in western Yunnan, southeastern Tibet Plateau; the meta-basalts were either emplaced or metamorphosed to greenschist facies at ∼30 Ma (^{40}Ar–^{39}Ar whole rock ages). A detailed rock magnetic study—in combination with reflected and transmitted light microscopy—identifies magnetite and Ti-rich titanomagnetite as the main magnetic carriers. Using alternating field demagnetization, we separated well-clustered characteristic remanent magnetizations. A fold test indicates a syn-folding remanence acquisition, supporting an Oligocene magnetization due to evidence for regional Eocene–Oligocene shortening. The tilt corrected overall site mean direction at 30 per cent of unfolding yields declination/inclination = 042.2°/47.0°, corresponding to a clockwise rotation of 35.1° ± 12.7° with respect to stable Eurasia. This denotes an average rotation rate of 1.17 ± 0.42° Myr^{-1} since ∼30 Ma, ranging at the lower limit of the present-day, GPS-derived rotation rates. We explain the clockwise rotation by tectonic escape of the Baoshan and the Lanping-Simao blocks along the Ailao Shan shear zone. With the onset of shearing along the Chong Shan and Gaoligong Shan shear zones, the Baoshan Block continued its southeastward escape, decoupled from the Lanping-Simao Block between these two major shear zones. Crustal flow could explain rotation and southward escape after ∼20 Ma.

Key words: Palaeomagnetism applied to tectonics; Rock and mineral magnetism; Thermochronology; Asia.

1 INTRODUCTION

The collision of India and Eurasia at ∼50 Ma (Klootwijk *et al.* 1992; Leech *et al.* 2005; Najman *et al.* 2010) and the ongoing convergence of the continental plates resulted in the formation of the Tibetan Plateau and the Himalayan mountain range, and caused significant shortening and strike-slip faulting in Southeastern Asia (Fig. 1). Tracing movements of Tibetan crust around the Eastern Himalayan Syntaxis (EHS) is a key issue to understand the geodynamic processes after the India–Eurasia collision. In the last decades, several models have been proposed based on field studies, analogue-material laboratory experiments and numerical thermomechanic modelling (e.g. Tapponnier *et al.* 1982; England & Houseman 1986; Peltzer & Tapponnier 1988; Beaumont *et al.* 2004). Two end-member models are widely considered, that is, the classic 'tectonic escape' model (Molnar & Tapponnier 1975; Tapponnier *et al.* 1982) and the 'crustal flow' model (e.g. Royden *et al.* 1997; Clark & Royden 2000). GPS data show a wide area of clockwise rotations around the EHS and southward surface motion in Yunnan (Fig. 1; Wang *et al.* 2001; Zhang *et al.* 2004; Gan *et al.* 2007; Sol *et al.* 2007; Banerjee *et al.* 2008; Maurin *et al.* 2010). Several major deformation zones, mainly the Gaoligong Shan (GSSZ), the Chong Shan (CSSZ) and the Ailao Shan Red–River (ASSZ) shear zones, accommodated the southeastward displacement of crust in this area (Fig. 1; e.g. Leloup *et al.* 1995; Akciz *et al.* 2008; Lin *et al.* 2009).

Several palaeomagnetic studies have been carried out in the last decades in Southeastern Asia; they indicate clockwise rotation of blocks in the wide area east of the EHS (e.g. Funahara *et al.* 1992, 1993; Chen *et al.* 1995; Sato *et al.* 2001, 2007; Tanaka *et al.* 2008;

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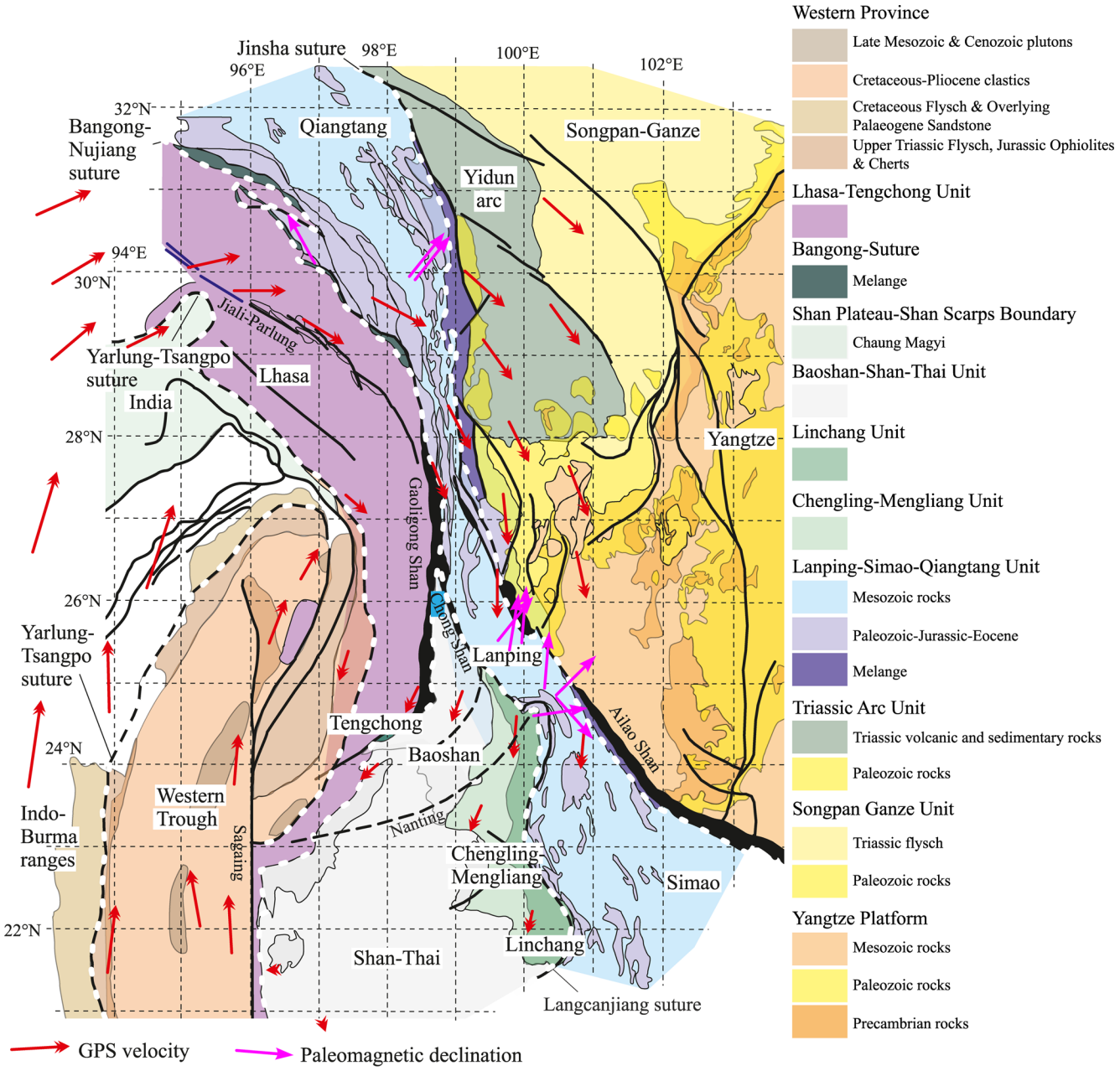


Figure 1. Simplified map of the Tibetan Plateau around the Eastern Himalayan Syntaxis (EHS). Colours denote the major crustal blocks of the EHS region, stippled lines sutures, black lines major fold zones and the three black bands the three major ductile shear zones of the region, that is, the Gaoligong, Chong Shan and Ailo Shan shear zones. The blue rectangle marks the study area. Present-day GPS velocities relative to Eurasia (Gan *et al.* 2007; Banerjee *et al.* 2008; Maurin *et al.* 2010) and selected palaeomagnetic declination data (Otofuji *et al.* 2010) are indicated by arrows.

Kondo *et al.* 2012; Tong *et al.* 2013). Rotations not consistent with this regional pattern were attributed to deformation within blocks. Palaeomagnetic studies near the EHS are rare. Huang & Opdyke (1991) and Ali *et al.* (2013) report data from Permian basalts of the Baoshan Block, reconstructing its former position within the Gondwana supercontinent.

Here, we present new palaeomagnetic data from meta-basalts of the northernmost Baoshan Block that acquired their magnetization at ~ 30 Ma. Samples stem from north of the town of Liuku along the Nujiang (Nu River; Fig. 2). These meta-basalt layers carry a syn-folding remanence and thus allow addressing rotations and translations accumulated in the late-stage history of the India–Asia collision.

2 GEOLOGICAL SETTING AND SAMPLING

Several microcontinents were amalgamated to the southern margin of Asia prior to the India–Asia collision (e.g. Yin & Harrison 2000; Metcalfe 2002, 2011). In the Tibetan Plateau these are from north to south the Kunlun, Qiangtang and Lhasa blocks. The latter continues into western Yunnan as the Tengchong Block (Fig. 1; e.g. Li *et al.* 2004; Xu *et al.* 2012). To the east, the Langcangjiang suture separates the Baoshan Block—the northern part of the Shan-Thai Block—from the Lanping-Simao Block, which is bounded by the ASSZ along its eastern margin (Fig. 1; e.g. Li *et al.* 2004; Akciz *et al.* 2008). The Gaoligong Shan (Gaoligong

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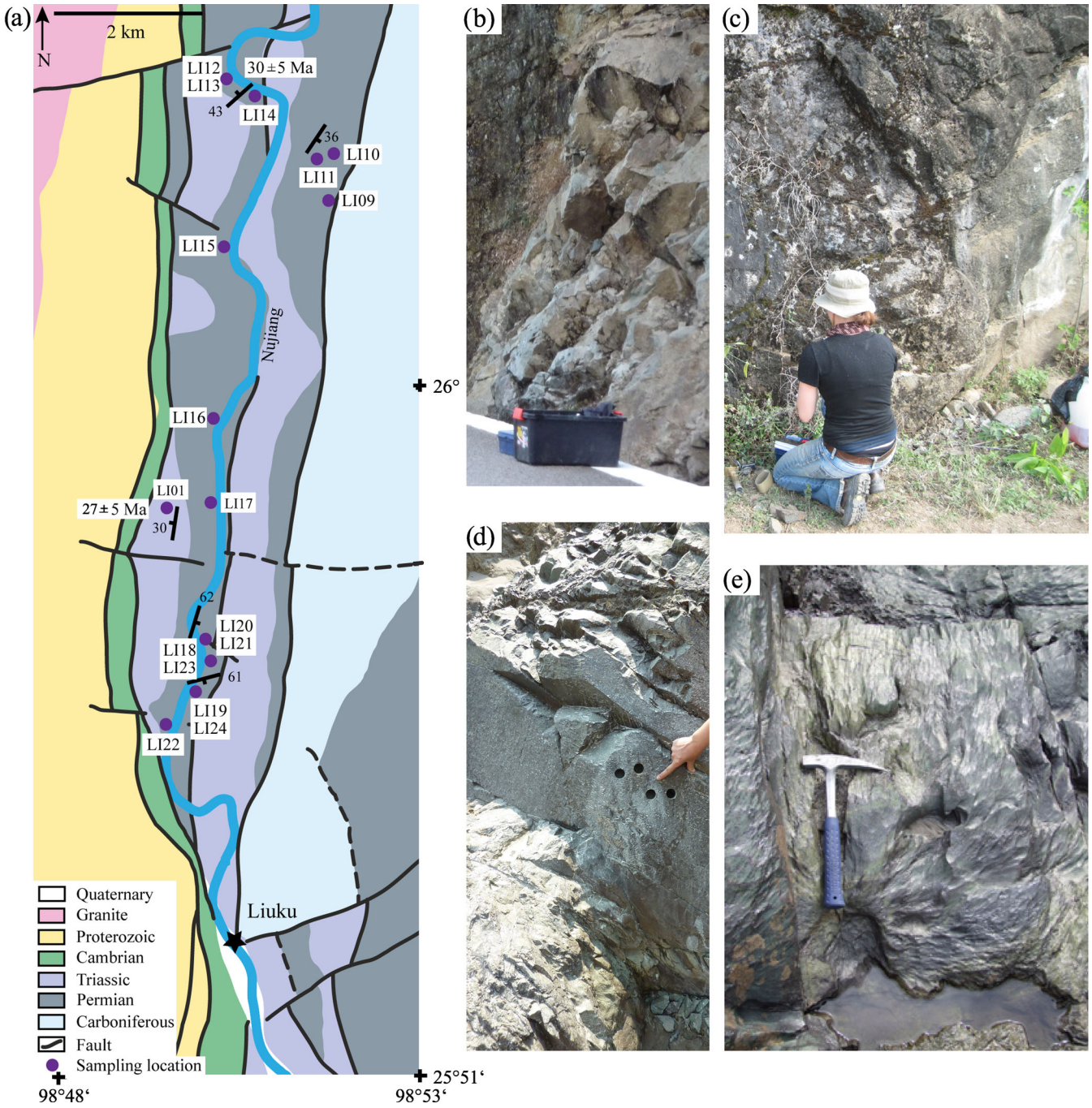


Figure 2. (a) Simplified geological map of the sampling area along the Nujiang (Nu River), showing the site locations of the meta-basalt layers (see Tables 1 and 2 for exact coordinates). Ages determined by our ^{40}Ar–^{39}Ar geochronology for sites LI01 and LI14 are shown in white boxes. (b–d) Field photographs of some of the sampled meta-basalt layers: (b) LI12, for the area west of the Nujiang; (c) LI11, (d) LI18 and (e) LI19 for the area east of the Nujiang.

Mountains), the Chong Shan and the Cenozoic Nanting fault zone confine the Baoshan Block into the west, east and south, respectively (Fig. 1; e.g. Socquet & Pubellier 2005). In the north, the Baoshan Block appears to pinch out where the GSSZ and CSSZ merge (Akciz 2004; Akciz *et al.* 2008). The Baoshan Block accreted to Asia in the Late Mesozoic shortened internally along N-trending folds and thrusts (present-day orientation) before or during the Eocene–Oligocene, possibly associated with the strike-slip movements along the GSSZ and CSSZ (Wang & Burchfiel 1997; Wang *et al.* 2008).

The Baoshan Block exposes thick Upper Proterozoic to Triassic marine strata, unconformably covered by Middle Jurassic marine and non-marine sedimentary rocks (e.g. Wang & Burchfiel 1997). The Palaeozoic strata are mainly carbonatic and clastic and locally intercalated with mainly rift-related tholeiitic basalts and basaltic volcaniclastic rocks that erupted when the Baoshan Block separated from Gondwana (e.g. Wopfner 1996; Wang *et al.* 2008). Rare Upper Eocene–Oligocene conglomerate and sandstone rest unconformably on older rocks and may have been deposited during the regional shortening and strike-slip deformation (e.g. Wang

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& Burchfiel 1997). Pliocene volcanic and sedimentary rocks and Quaternary strata remain mostly undeformed (Wang & Burchfiel 1997). The sampled meta-basalts either belong to the Permian Woniusi Formation (e.g. Ueno 2003)—observed in the southern Baoshan Block, characterized by a cold/cool-water fauna (e.g. Metcalfe 2002; Ueno 2003), and not described in our sampling area—or are part of the ∼40–30 Ma potassic volcanic rocks of Yunnan and northern Vietnam, which are likely related to Late Eocene–Oligocene intraplate extension associated with the India–Eurasia collision (e.g. Chung *et al.* 1997, 2005).

17 meta-basalt sites were sampled with a portable rockdrill along the Nujiang valley north of Liuku for a palaeomagnetic study (Fig. 2). Each site consists of eight cores with 2.5 cm diameter; we cut the cores into ∼2.3-cm-long specimens. The orientation of the cores was measured with a magnetic compass and an inclinometer. In order to attempt a tilt correction, we measured the layer dips of the meta-basalts layers and the bedding of the limestones that are in contact with the meta-basalts; the layer dips of sites LI15, 16, 17, 22—where direct dip measurements were impossible—were taken from the nearest neighbouring sites. Bedding strikes ∼N throughout the study area (Fig. 2; Socquet & Pubellier 2005).

3 MAGNETIC MINERALOGY

3.1 Analytical methods

To determine the ferro(i)magnetic mineralogy, we used one specimen per site. Isothermal remanent magnetization (IRM) curves were obtained using a pulse magnetizer MMPM9 with a maximum field of 2.5 T; intensity was measured using a spinner magnetometer (MINISPIN-Molspin). The IRM acquisition curves were processed by cumulative log-Gaussian (CLG) distributions using the software irmunmix2_2_1 and IRM_CLG1 (Kruiver *et al.* 2001). Afterwards, the saturation IRM (SIRM) was stepwise (25–50 °C) thermally demagnetized up to a maximum temperature of 700°C. Heating was done using an ASC furnace and a MMTD-18 furnace. High and low temperature thermomagnetic runs were performed on one powdered sample of each site using a Kappabridge KLY-3 (Agico, Brno, Czech Republik) with a CS-3 temperature unit. Bulk magnetic susceptibility of each specimen was measured using a Kappabridge KLY-2 (Agico). Hysteresis loops were measured with a Princeton Measurements 2900 MicroMag (LakeShore, Westerville, Ohio, USA) alternating gradient force magnetometer (AGFM).

Light microscopy analysis was accomplished using a Leica DM 2500P microscope (Leica, Wetzlar, Germany) with a 20×/0.4 oil objective for reflected light microscopy. Energy dispersive X-ray (EDX) analysis in combination with scanning electron microscopy (SEM) was done using an EDX-system OXFORD INCA Energy 200 Premium Si (Li) SATW-Detector (Oxford Inca, Abingdon, Oxfordshire, United Kingdom) and an SEM LEO Model 1450 VP.

3.2 Microscopy results

Five polished thin sections (sites LI01, LI09, LI10, LI14 and LI20) were prepared for transmitted-light microscopy and eight polished thin sections (sites LI01, LI09, LI10, LI14, LI15, LI17, LI20 and LI22) for reflected-light microscopy.

At sites LI09 and LI10 coarse-grained carbonate (possibly siderite), pistazite, chlorite, limonite and goethite occur as phenocrysts in the matrix (Figs 3a–c). The latter is highly altered with identifiable plagioclase and minor carbonate. Haematite occurs along the rims, and as inclusions in plagioclase, as fracture fillings and partly formed by martitisation (Figs 3a–c). No carbonate was observed in the other specimens. The matrix of site LI01 is fine-grained (Fig. 3d); it is coarse-grained in the other meta-basalts. LI01 (Fig. 3d)—without phenocrysts—contains small plagioclase needles, chlorite, epidote and anatase/titanite. Sites LI14 and LI20 contain partly large plagioclase phenocrysts and plagioclase, and actinolite, epidote and chlorite in the matrix; relatively coarse-grained epidote and locally chlorite fill vesicles (Fig. 3e). At sites LI14 and LI22, we identified goethite, chalcopyrite and anatase/titanite (Fig. 3f). Sites LI15 and LI20 additionally contain haematite and skeleton-shaped (titano-)magnetite (Fig. 3g), typical for rapidly cooled basalts (Petersen & Vali 1990).

All samples show strong hydration but the original igneous texture is preserved. The assemblage epidote + actinolite + chlorite + albite and the lack of prehnite and pumpellyite suggests lower greenschist–facies metamorphism, likely above 250–300 °C. The presence of titanomagnetite—if metamorphic—suggests temperatures above 345–385 °C (Yang 1987). Iron oxyhydroxides (chalcopyrite and goethite) indicate late-stage weathering.

3.3 Results of rock magnetic analysis

Thermal demagnetization (ThD) of SIRM (Fig. 4a) reveals the presence of magnetite and haematite by a decrease in magnetization at ∼580 °C (in all except two samples) and above 650 °C (in eight samples), respectively. For most samples (except LI09, LI12 and LI24), Ti-rich titanomagnetite is additionally indicated by a decrease in magnetization between 200 and 500 °C. The varying range in unblocking temperatures is explained by different concentrations of Ti and other cations in the crystal lattice of titanomagnetite and by a variable degree of low-temperature oxidation (LTO; Dunlop & Özdemir 1997). In samples LI09 and LI12, only the presence of haematite is indicated.

IRM acquisition curves saturate at fields around 300 mT and higher (Fig. 4a); only sample LI09 (containing haematite) does not reach saturation. At four sites, CLG-analysis (Fig. 4a) indicates the presence of two ferrimagnetic phases. The lower coercive component (LCC; $B_{1/2}$ = 38–97 mT) likely represents a relatively soft coercive magnetite fraction and the relatively harder fraction ($B_{1/2}$ = 126–203 mT) is likely representing Ti-rich titanomagnetite. These relatively high coercivities are explained by irregular internal stresses due to magnetostriction that generally dominates the internal anisotropy of natural Ti-rich titanomagnetites (Appel & Soffel 1984; Appel 1987). Samples with one component (12 sites) show $B_{1/2}$ values of 59–122 mT, indicating either magnetite or Ti-rich titanomagnetite. Six samples also contain a small amount of haematite ($B_{1/2}$ = 630–1577 mT), but its contribution to the SIRM does not exceed 10 per cent.

High-temperature thermomagnetic runs of magnetic susceptibility (κ–T curves) confirm the presence of one or two ferro(i)magnetic phases (Fig. 4b). A Curie temperature near 580 °C, identifying magnetite, is observed for all measured samples. The Verwey transition of magnetite (Verwey 1939) at ∼120 K further implies its occurrence (Fig. 4b). A peak in the low temperature κ–T curves—slightly above the Verwey transition (Fig. 4b)—can be related to the isotropic point of magnetocrystalline anisotropy (Syono 1965). Its pronounced appearance can be interpreted by a dominance of multidomain (MD) particles (Moskowitz *et al.* 1998). Results from hysteresis loop measurements also indicate the presence of MD and PSD (pseudo-single domain) particles and a mixture of both

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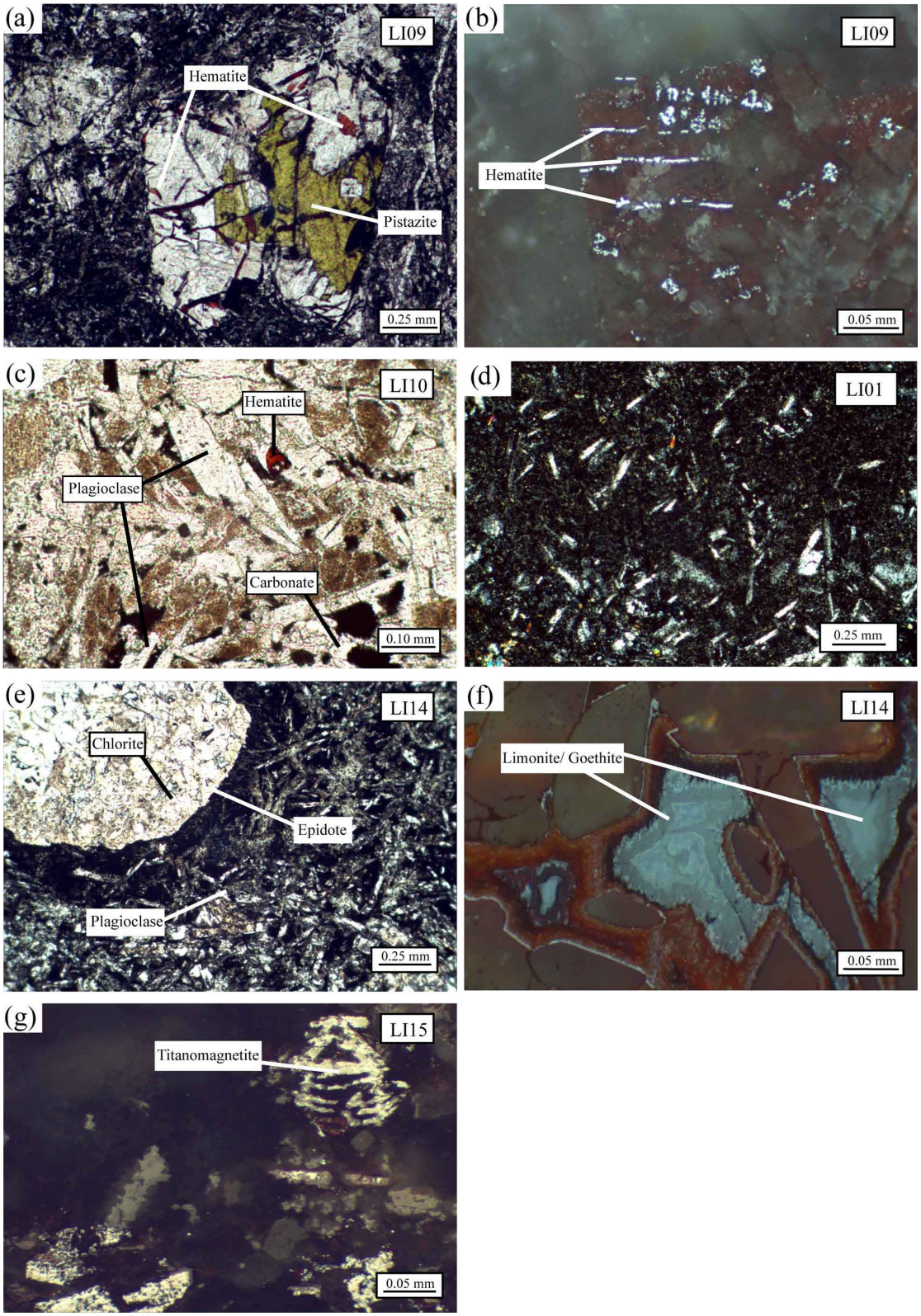


Figure 3. (a) Sample LI09: transmitted light photograph of pistazite phenocryst and haematite; (b) shows haematite in reflected light. (c–e) Samples LI10, LI01 and LI14: transmitted light photographs of matrix mineralogy and large epidote aggregates filling a vesicle. (f, g) Samples LI14 and LI15: reflected light photographs of limonite/goethite phenocrysts and a large skeletal titanomagnetite.

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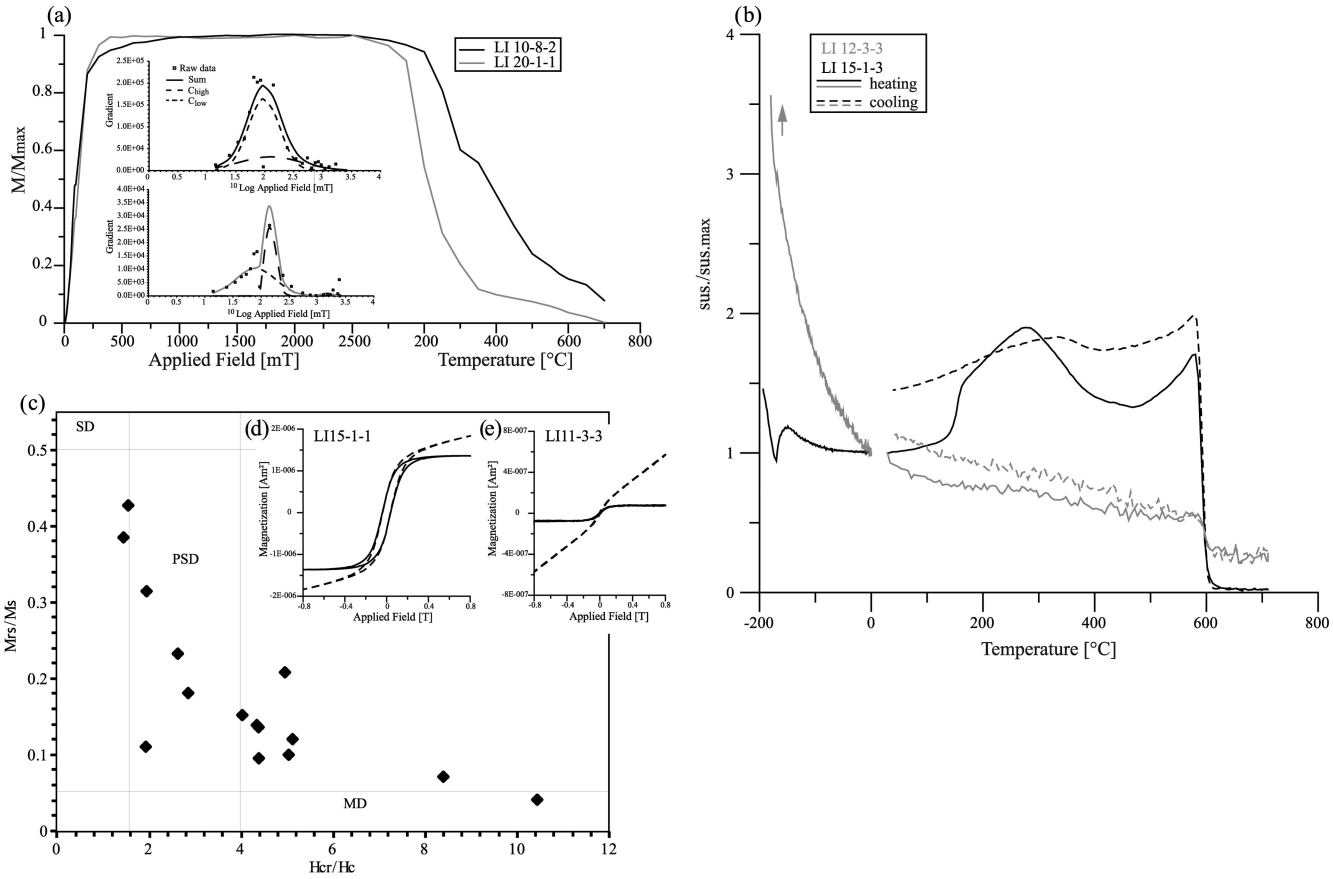


Figure 4. (a) IRM acquisition curves and thermal demagnetization of SIRM for representative specimens. Results of cumulative log-Gaussian analysis (Kruiver *et al.*, 2001) are shown as insets (c_{low}: lower coercivity component, c_{high}: higher coercivity component). Magnetic susceptibility versus temperature curves of rock powders are shown by (b) low-temperature and high-temperature curves on the left- and right-hand side, respectively (heating: full line, cooling: dashed line). (c) Day-plot (Day *et al.* 1977, Dunlop & Özdemir 1997) of one sample from each site. (d, e) Hysteresis loops of two representative specimens (LI15-1-1, LI11-3-3); dashed line, no correction; solid line, corrected for paramagnetic fraction at 70 per cent of H_{max}; M_{rs}, saturation remanence; M_s, saturation magnetization; H_{cr}, remanent coercivity; H_c, coercivity; SD, single domain; PSD, pseudo-single domain; MD, multidomain.

(Figs 4c–e). Five samples show a second decrease in the κ–T heating curves at intermediate temperatures, which is only weakly represented in the cooling curves. In conjunction with the ThD results of SIRM, the most likely interpretation is the presence of Ti-rich titanomagnetite that is almost completely destroyed at higher temperatures or exsolve into magnetite- and ilmenite-near phases by high-temperature oxidation (Moskowitz 1981).

4 ^{40}AR–^{39}AR GEOCHRONOLOGY

We analysed two whole-rock meta-basalts by the ^{40}Ar–^{39}Ar method (samples LI01, LI14; Fig. 5; Tables 1 and S1). The Appendix gives a summary of the hardware and analytic protocols utilized at the TU Bergakademie Freiberg Ar–Ar facility (ALF). In thin section, the two meta-basalts show volcanic textures mostly traced by plagioclase. A low-grade metamorphic mineral assemblage, comprising variable amounts of epidote, chlorite, actinolite and anatase/titanite, traces various stages of hydration. According to this overprint, the Ar-release pattern of the meta-basalts is complex and hampers a straightforward interpretation of the ^{40}Ar–^{39}Ar data (Fig. 5 and Table 1).

Meta-basalt LI01 yielded apparent ages between ∼45 and 5 Ma. Four data points can be fitted by an isochron with a—within error—initial ^{40}Ar–^{36}Ar composition close to the atmospheric value (295.5). We speculate that the approximate age of metamorphism of the meta-basalt is at 27 Myr. The apparent ages of meta-basalt LI14 cover ∼65–25 Myr; the first 20 steps and four intermediate temperature steps define isochrons with about atmospheric ^{40}Ar–^{39}Ar composition that suggest basalt metamorphism at ∼33 Ma. Given the occurrence of regional Cenozoic (∼45–25 Ma) magmatism and thus regional crustal heating across western Yunnan (see compilation in Kornfeld *et al.*, in preparation), we interpret that the Nujiang meta-basalts trace a thermal event at ∼30 Ma (±5 Ma); however, the strong hydration precludes a more precise age assignment. Note that it is unimportant for the interpretations put forward in this paper whether the remanence was acquired in the basalt layers during their emplacement or by the metamorphic formation of the magnetic carriers, as a result of regional heating due to crustal shortening and magmatism. The important issue is that the magnetic carriers formed or were reset at ∼30 Ma.

5 PALAEOMAGNETIC ANALYSES

5.1 Methods

Twin specimens from each site were chosen for pilot demagnetization analyses; we progressively demagnetized one of them in an

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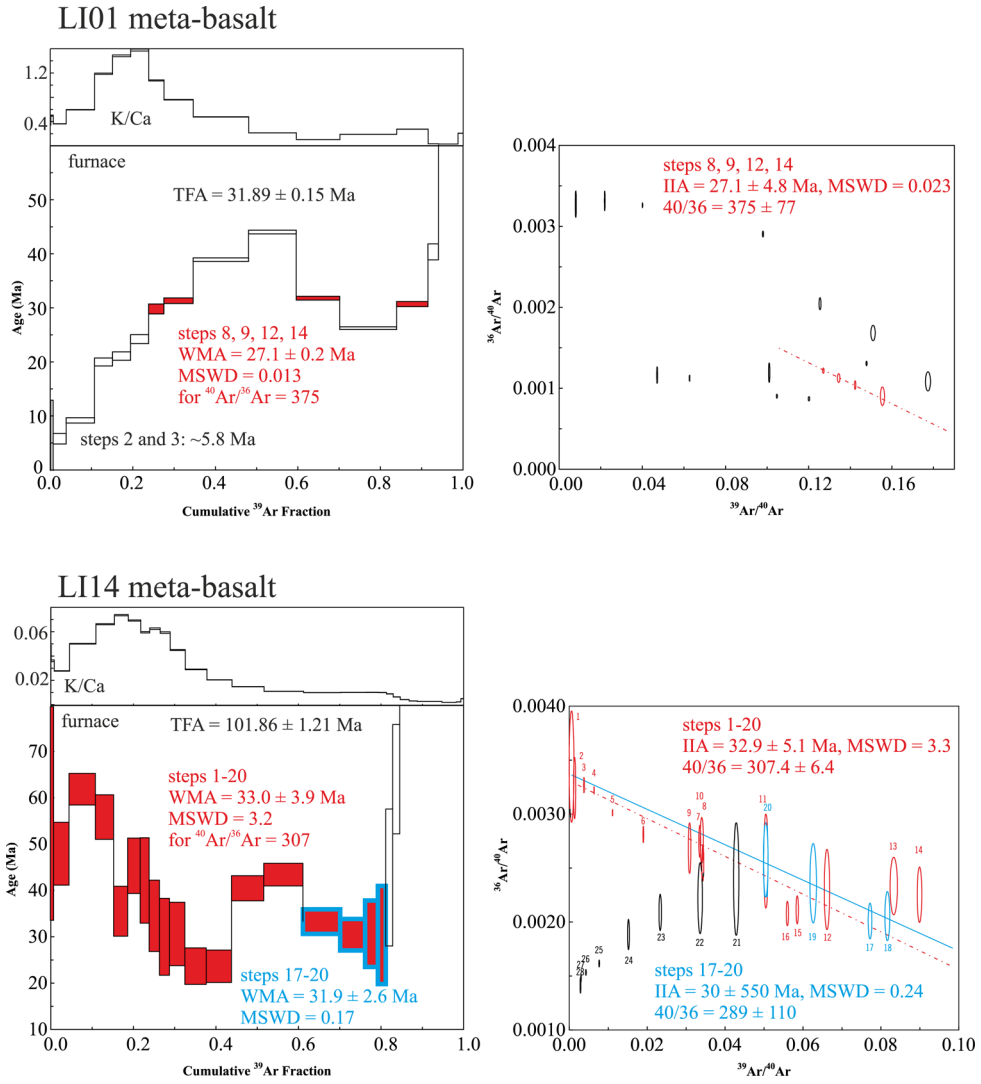


Figure 5. ^{40}Ar–^{39}Ar thermochronologic results of meta-basalts of the Nu River section, displayed as plateau age spectra and corresponding inverse isochron diagrams.

Table 1. Summary of ^{40}Ar/^{39}Ar data.

Sample	Longitude (°E)	Latitude (°N)	Elevation (m)	Rock	mg	TFA (Ma)	WMA (Ma)	Isochron (invers) Ma	MSWD	40/36	Per cent Ar	Steps
LI01	98.8533	26.0664	886	Basalt	25.03	31.89 ± 0.15	27.1 ± 0.2 (40/36 = 375)	27.1 ± 4.8	0.023	375 ± 77	28.8	8, 9, 12, 14 (17)
LI14	98.83	25.9658	1353	Basalt	55.7	101.86 ± 1.21	31.9 ± 2.6	30 ± 550	0.24	289 ± 110	20.1	17–20 (20)

Notes: MSWD is the mean square weighted deviation, which expresses the goodness of fit of the isochron. Isochron and weighted mean ages (WMA) are based on fraction of ^{39}Ar and steps listed. Preferred age interpretation: LI01 meta-basalt, 27 ± 5 Ma, probably overprint at ~6 Ma; LI14 meta-basalt, 31 ± 5 Ma.

alternating field and the other thermally. Remanence directions were measured with a 2G Enterprises (CA, USA) He-free SQUID magnetometer 755–1.65 UC (sensitivity limit 10^{-7} Am^{-1} for a specimen volume of 10 cm^3). For alternating field demagnetization (AF) a 2G automatic degaussing system integrated in the SQUID magnetometer was used. Heating was performed using an ASC furnace and an MMTD-18 furnace. Steps of 2–20 mT were chosen for AF with a maximum field of 100 mT. ThD was carried out using steps of 25–50 °C with a maximum temperature of 650 °C. Magnetic susceptibility was measured using a Kappabridge KLY-2 from Agico after each heating step in order to detect possible magneto-mineralogical changes. For separation of remanent magnetization components AF turned out to be the better option [demagnetization paths smoother than for ThD (Figs 6f and h); almost full demagnetization reached] and it was therefore applied for the remaining specimens. In total eight specimens were measured from each site.

5.2 Palaeomagnetic results

Natural remanent magnetization (NRM) intensity values are quite heterogeneous between different sites and partially also among the samples within a site, ranging between 0.9 and 161 m Am^{-1} (Table 2). Four sites yield NRM values above 100 m Am^{-1}, three

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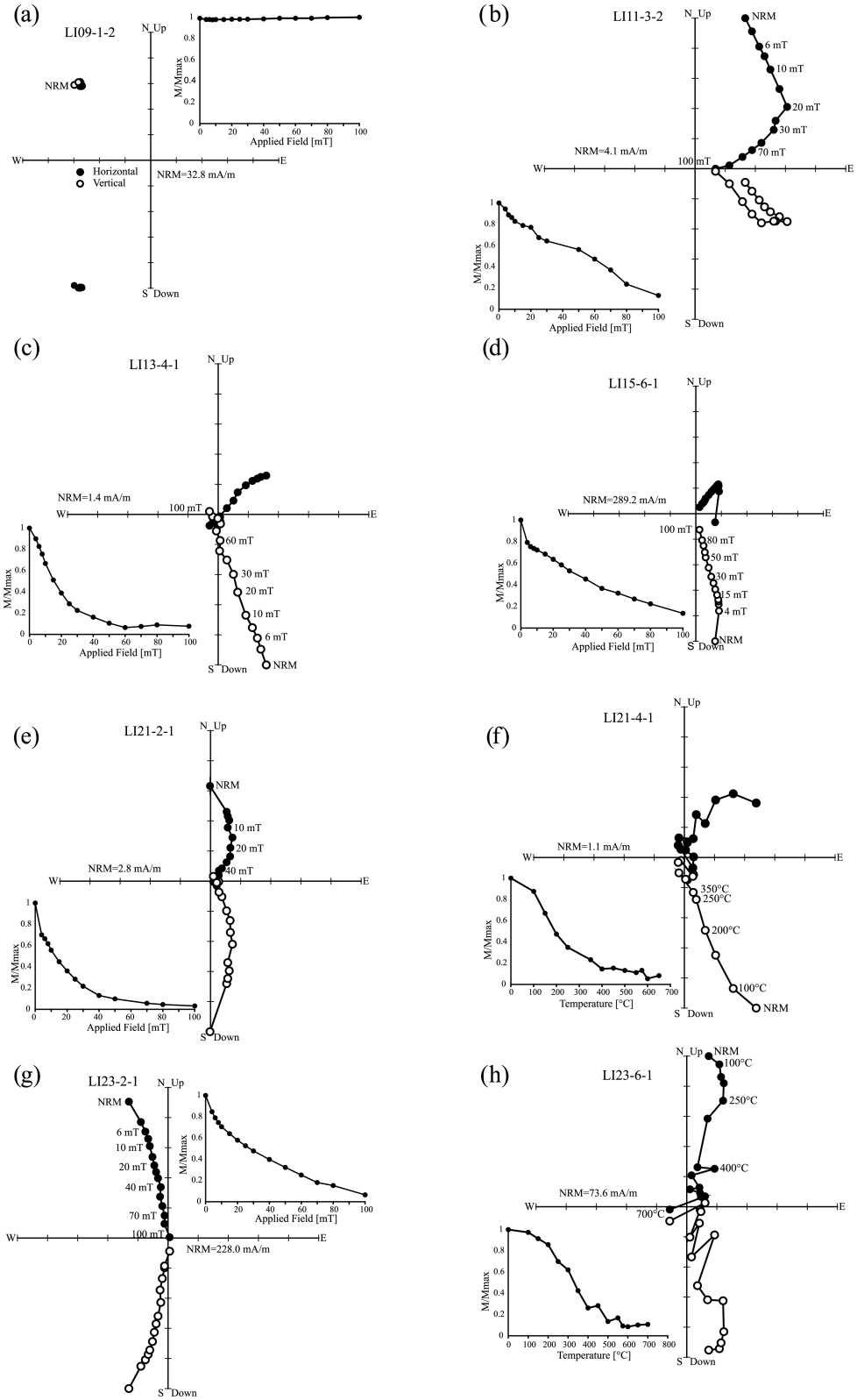


Figure 6. Zijderveld diagrams (in geographic coordinates) and intensity curves for alternating field demagnetization (a–e and g) and for thermal demagnetization (f, h) of representative specimens.

sites around 20 m Am^{-1} and the remaining sites have intensities of less than 10 m Am^{-1}. For sites with relatively higher NRM and susceptibility values, we generally observed Ti-rich titanomagnetite, whereas most sites with low NRM and susceptibility values only contain magnetite (and haematite). The low NRM values may be caused by oxidation of Ti-rich titanomagnetite that transforms to magnetite (Prévot *et al.* 1981) and partly further to weakly magnetic haematite or non-ferrimagnetic phases.

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Table 2. Site mean values and statistical parameters of the palaeomagnetic results.

Site	Site location Lat.	Long.	Strike/Dip	NRM (mA m^{-1})	κ (10^{-6} SI)	NRM/κ (A m^{-1})	C	*In situ* Dec. (°)	*In situ* Inc. (°)	Tilt-corrected Dec. (°)	Tilt-corrected Inc. (°)	k	α_{95} (°)	N/N incl. (n/r)	IRM Comp. 1 $B_{1/2}$ (mT)	IRM Contrib. (per cent)	IRM Comp. 2 $B_{1/2}$ (mT)	IRM Contrib. (per cent)
LI 01	25.9658	98.8300	**190/30**	6.0	926	6.5	LCC	300.6	27.7	274.0	3.3	2.4	55.7	6/6 (6/0)				
							HCC	55.1	29.4	55.8	30.4	9.8	22.5	6/6 (4/2)	66.5	100		
LI 09	26.0403	98.8631		66.0	622	106.1									80.0	32	747.6	68
LI 10	26.0464	98.8667	**234/36**	159.3	2332	68.3	LCC	356.9	34.3	349.7	15.0	6.6	32.2	5/5 (5/0)				
							HCC	307.1	43.3	309.4	32.7	4.0	34.6	7/7 (7/0)	96.6	72	126.1	28
LI 11	26.0467	98.8645	**234/36**	6.6	927	7.1	LCC	331.8	-6.0	332.2	-16.9	4.9	27.8	8/8 (8/0)				
							HCC	60.4	26.5	54.8	27.3	24.6	13.8	8/6 (6/0)	59.0	99[a]		
LI 12	26.0718	98.8444	**335/43**	6.1	800	7.6	LCC	349.3	40.0	358.9	35.8	12.9	22.1	5/5 (3/2)				
							HCC	60.3	61.0	61.6	48.0	227.1	5.1	8/5 (5/0)	38.1	89	202.9	11
LI 13	26.0718	98.8444	**335/43**	3.4	610	5.6	LCC	323.8	2.0	324.5	4.4	3.1	45.7	6/6(5/1)				
							HCC	36.0	60.5	43.8	48.7	44.5	8.4	8/8 (8/0)	82.1	100		
LI 14	26.0664	98.8533	**335/43**	124.1	10720	11.6	LCC	16.6	33.7	21.8	24.6	8.0	25.3	7/6 (5/1)				
							HCC	36.7	53.8	42.8	42.0	46.4	8.2	8/8 (8/0)	63.2	100		
LI 15	26.0279	98.8450	335/43	161.1	6408	25.1	LCC	15.8	57.4	22.3	41.6	23.2	19.5	6/4 (3/1)				
							HCC	40.8	72.0	50.4	59.7	344.5	4.1	8/5 (5/0)	80.3	61	156.9	39
LI 16	25.9866	98.8404	335/43	6.8	1084	6.3	LCC	340.5	−9.7	341.2	−7.4	4.1	34.0	8/7 (7/0)				
							HCC	25.9	67.0	39.5	56.1	39.6	10.8	8/6 (6/0)	67.4	99[a]		
LI 17	25.9687	98.8410	335/43	17.9	1182	15.1	LCC	5.8	50.0	350.0	55.1	40.1	10.7	8/6 (6/0)				
							HCC	226.6	66.4	232.6	53.9	44.5	8.4	8/8 (8/0)	79.5	100		
LI 18	25.9325	98.8337	**69/61**	23.0	1170	19.7	LCC	10.8	14.0	14.7	29.2	5.3	36.1	7/5 (5/0)				
							HCC	343.0	58.7	346.7	76.6	15.9	15.6	7/7 (7/0)	81.5	100		
LI 19	25.9357	98.8365	**69/61**	3.3	706	4.7	LCC	13.2	6.7	15.9	21.4	4.5	48.7	5/4 (4/0)				
							HCC	8.3	41.3	20.2	56.1	97.3	6.8	7/6 (6/0)	67.2	100		
LI 20	25.9431	98.8367	**30/62**	17.6	1644	10.7	HCC	19.3	41.6	36.5	42.2	130.9	4.9	8/8 (8/0)	79.8	58	155.7	40[a]
LI 21	25.9431	98.8367	**30/62**	1.1	646	1.7	LCC	315.2	57.9	333.7	75.4	16.7	31.1	3/3 (3/0)				
							HCC	20.6	34.5	34.0	35.4	22.8	12.9	7/7 (7/0)	62.0	93[a]		
LI 22	25.9258	98.8288	335/43	0.9	619	1.5	LCC	49.8	62.6	345.1	16.7	2.2		4/2 (2/0)				
							HCC	30.7	68.6	42.8	57.1	32.6	13.6	7/5 (5/0)	60.2	90[a]		
LI 23	25.9325	98.8337	**69/61**	140.4	3873	36.3	LCC	343.4	27.2	351.9	55.4	87.4	9.9	5/4 (4/0)				
							HCC	0.6	43.8	14.5	51.5	28.1	14.7	8/5 (5/0)	121.6	100		
LI 24	25.9357	98.8365	**69/61**	9.4	906	10.4	LCC	350.6	49.5	341.3	63.1	3.8	35.6	8/7 (7/0)				
							HCC	261.3	68.9	217.4	65.7	271.7	3.4	8/8 (8/0)	67.7	99[a]		

Note: Strike/Dip values in bold were used for fold test.

[a]Samples may contain up to 10 per cent of a higher coercivity phase ($B_{1/2}>1$T).

Lat., latitude; Long., longitude; NRM, mean natural remanent magnetization; κ, susceptibility; C, components (LCC—low-coercivity comp., HCC—high-coercivity comp.); Dec., declination; Inc., inclination; k: precision parameter; α_{95}, 95 per cent confidence angle; N/N incl. (n/r)., number of specimens measured/included in statistics (normal/reverse polarity); IRM, isothermal remanent magnetization (decomposed by cumulative log-Gaussian distributions after Kruiver *et al.*, 2001); Comp., component; Contrib., contribution.

Table 3. Overall site mean directions.

	Dec. (°)	Inc. (°)	k	α_{95} (°)	α_{95}/cos I (°)	N incl.
Mean LCC:	346.2	30.1	7.5	15.6	18.0	14
Mean HCC (excl. LI10, LI17, LI18, LI24):	32.6	51.8	15.7	11.3	18.3	12
Mean HCC tilt-corrected (excl. LI10, LI17, LI18, LI24):	42.2	47.0	32.2	7.8	11.4	12

Dec., declination; Inc., inclination; k, precision parameter; α_{95}, 95 per cent confidence angle; α_{95}/cos I, 95 per cent confidence limit of the declination; N incl., number of specimens included in statistics.

In most samples, the AF curves show a stable demagnetization path. The AF did not work for site LI09, as it contains high amounts of haematite. Therefore, we excluded this site from further analyses (Fig. 6a). An LCC and a higher coercive component (HCC) was separated at field intervals of about 0–30 mT and 4–100 mT (Figs 6b–e and g), respectively, using principal component analysis (Kirschvink 1980). Site means were calculated using Fisher statistics (Fisher 1953). Within sites, specimen directions show good grouping for the HCC; the LCC is relatively scattered. The heterogeneous NRM distribution reveals no significant effect with respect to the grouping of site mean directions or overall mean directions. The statistical results are listed in Table 2.

The LCC *in situ* overall site mean directions [Fig. 7a; Table 3; declination/inclination (D/I) = 346.2°/30.1°; k = 7.5, α_{95} = 15.6°; N = 14] plots close to the present-day Earth magnetic dipole field (D/I = 0°/44.2°). It likely represents a younger overprint and will not be discussed further. The *in situ* HCC are mostly well grouped within sites (generally $k > 15$ and mostly much better; Table 2); they yielded an overall mean direction (Table 3) of D/I = 32.6°/51.8° (k = 15.7, α_{95} = 11.3°; N = 12). Four sites (LI10, LI17, LI18 and LI24; grey dots in Figs 7b and c) have been excluded from the overall mean direction. These sites show distinctly divergent and widely scattered directions. The reason for this scatter is unclear; we cannot exclude that these sites were drilled in landslide blocks, misidentified as outcrops due to the dense vegetation cover. The HCC direction of site LI01 shows mixed polarities (Fig. 8), whereas in the other sites only normal polarity directions occur. The mixed polarities indicate that the time of remanence acquisition spanned a longer period, at least comprising one reversal of the Earth's magnetic field. However, because of the occurrence of more than one ferrimagnetic phase a self-reversal mechanism cannot be excluded (Krása *et al.* 2005).

A fold test was performed for sites where the layering could be directly measured (Table 2). The direction-correction fold test (Enkin 2003) is indeterminate, yielding an optimal degree of untilting at 34.4 ± 15.2 per cent at the 95 per cent confidence limit. The parametric simulation fold test (Watson & Enkin 1993) yields an optimal degree of untilting at 30 per cent of unfolding (Fig. 7d),

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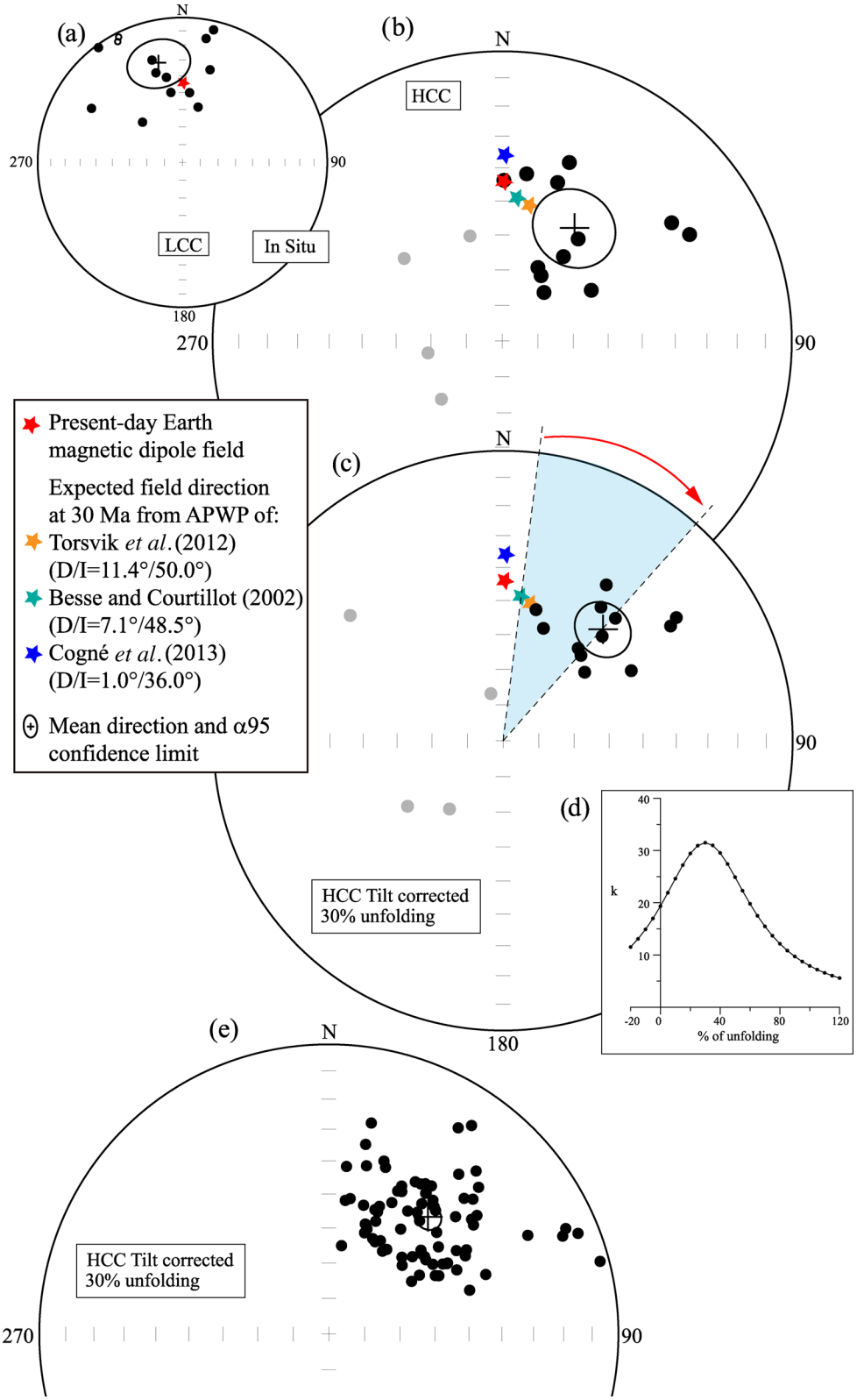


Figure 7. Equal area projection in geographic coordinates of site mean directions of (a) the LCC and (b) the *in situ* HCC. (c) Tilt-corrected (30 per cent unfolding) HCC mean directions. (d) Results from the parametric simulation fold test (Watson & Enkin, 1993); light grey symbols: outliers excluded from statistics. (e) Tilt-corrected (30 per cent unfolding) HCC specimen directions.

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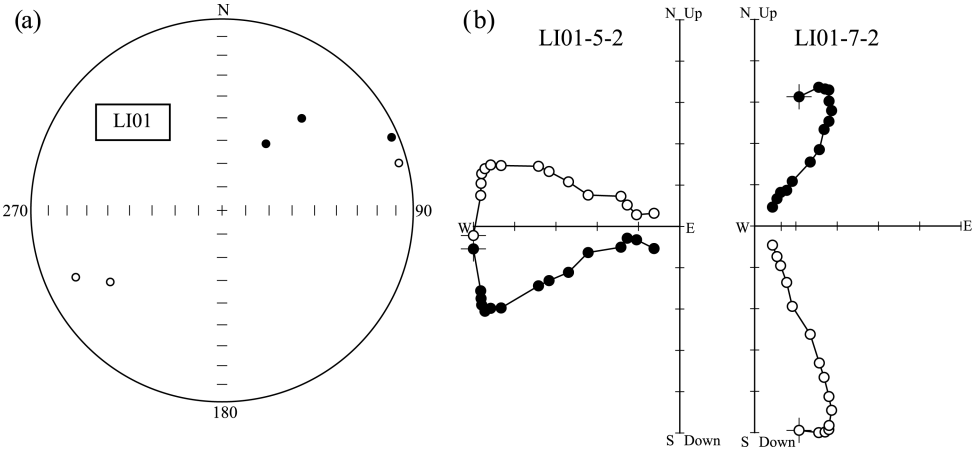


Figure 8. (a) Equal area projection of antipodal *in situ* HCC single-specimen directions for site LI01 (full/open circles: positive/negative inclination). (b) Zijderveld diagrams of samples LI01–5–2 (reverse) and LI01-7-2 (normal).

indicating a syn-folding remanence acquisition. Thus, we performed a 30 per cent untilting of the HCC mean direction. The tilt-corrected HCC site mean directions (Fig. 7c) are well grouped; the 30 per cent untilted overall mean direction is $D/I = 42.2^\circ/47.0^\circ$ ($k = 32.2$, $\alpha_{95} = 7.8^\circ$; $N = 12$; four sites excluded; Table 3). Excluding sites where layering was inferred from neighbouring sites (LI15, LI16, LI17 and LI22) yields a similar result: $D/I = 41.8^\circ/43.4^\circ$ ($k = 29.9$, $\alpha_{95} = 9.6^\circ$; $N = 8$).

The age of remanence acquisition is a crucial issue. Our HCC site mean directions are clearly not in agreement with a Permian age (Huang & Opdyke 1991; Ali *et al.* 2013). However, they coincide with the Eocene–Oligocene shortening within the Baoshan Block (Wang *et al.* 2008). Our HCC results show no difference between the remanence directions carried by magnetite and Ti-rich titanomagnetite, and the latter has definitely formed during cooling of the basalts. We did not observe any signs of high-temperature oxidation (typically expressed by magnetite and ilmenite lamellae; e.g. Dunlop & Özdemir 1997) in our microscopic investigations, thus the origin of magnetite must be related to LTO. As LTO is a process of alteration that is enhanced at elevated temperatures, metamorphism is the most likely process where magnetite was predominately formed. Consequently, we can assume an ~30 Ma HCC remanence acquisition age, independent whether our thermochronologic result represents an emplacement or metamorphic age.

To test whether palaeosecular variation (PSV) is averaged out, we first determined the mean remanence direction of the HCC of all specimens. A mean value for the bedding-corrected results of $D/I = 40.2^\circ/46.0^\circ$ ($N = 79$, $k = 19.6$, $\alpha_{95} = 3.7^\circ$; Fig. 7e) was achieved. The precision parameter k' (McFadden 1980) at the latitude of our sampling sites (26°N) is $k' = 28.5$. Comparing the k-value from the mean of all our specimens and the site mean directions with k', our k value is mostly below k', thus PSV is represented in the HCC. This slightly higher scatter can be explained by imperfect remanence recording and/or some degree of remanence alteration.

6 TECTONIC IMPLICATIONS

In order to determine the amount of block rotation, we compared the overall tilt-corrected HCC mean direction ($D/I = 42.2^\circ/47.0^\circ$) with the expected direction for the 30 Ma Eurasian palaeopole. Using the apparent polar wander path (APWP) of Besse & Courtillot (2002) (expected $D/I = 7.1^\circ/48.5^\circ$; $\alpha_{95} = 3.8^\circ$; $N = 23$) yields a clockwise rotation of $35.1^\circ \pm 12.7^\circ$ (Fig. 7c) with respect to stable Eurasia, corresponding to a rotation rate of $1.17 \pm 0.42^\circ$ Myr^{-1}. In comparison with the recently proposed new APWP for East Asia by Cogné *et al.* (2013) (expected $D/I = 2.1^\circ/36.4^\circ$; $\alpha_{95} = 2.8^\circ$; $N = 25$), the magnitude of clockwise rotation ($40.1^\circ \pm 11.9^\circ$) and the corresponding rotation rate ($1.34 \pm 0.40^\circ$ Myr^{-1}) remain similar. However, the expected inclination value is significantly different, and thus the expected palaeolatitude. Taking Torsvik *et al.*'s (2012) APWP into account, we obtain a similar rotation (30.8°) but the large α_{95} confidence limit (21.1°) for 30 Ma (due to only $N = 2$ data) leads to a very high uncertainty (34.7°). All confidence limits for rotation values were calculated from ΔD_{obs} and ΔD_{exp} (95 per cent confidence angles of the observed and expected declinations: $\Delta D = \alpha_{95}/\cos I$) using the error propagation law.

Tapponnier *et al.*'s (1982) plasticine experiments suggested an ~40° clockwise rotation of the Shan–Thai Block, which was supported by subsequent palaeomagnetic studies, performed mostly on Palaeozoic to Mesozoic rocks (e.g. Funahara *et al.* 1993; Chen *et al.* 1995; Sato *et al.* 2001, 2007; Tanaka *et al.* 2008). Only the study of Chen *et al.* (1995) included data from Palaeogene to Neogene strata. Our $35.1^\circ \pm 12.7^\circ$ clockwise rotation of the Baoshan Block is in line with their data and therefore resolves that the full magnitude of rotation occurred during the period since ~30 Ma. The average rotation rate of $1.17 \pm 0.42^\circ$ Myr^{-1} is at the lower limit of the rates of 1.14–2.40° Myr^{-1} around the EHS reported by Otofuji *et al.* (2010) from present-day GPS velocities and Quaternary strain rates (Holt *et al.* 2000).

In conjunction with rotation, southeastward translation of lithospheric blocks has been proposed (e.g. Tapponnier *et al.* 1982). The observed (28.2°N) and expected (29.5°N) palaeolatitudes reveal no significant difference within the 95 per cent confidence limit (dp) of the corresponding pole (6.5°) resulting from the overall mean direction. In addition the present-day latitudinal position of our sampling area (26°N) and the observed palaeolatitude are not statistically different. The small difference (2.2°) between the present-day position and the observed palaeolatitude corresponds to a southward translation of ~240 km. A southward displacement of the Baoshan Block is therefore possible but within the 95 per cent uncertainty range, the estimated translation is insignificant.

In contrast, the APWP of Cogné *et al.* (2013) yields an expected palaeolatitude of 20.2 ± 1.9°N, leading to a latitudinal difference of $8.0^\circ \pm 6.8^\circ$ with respect to the observed one (28.2°N). This

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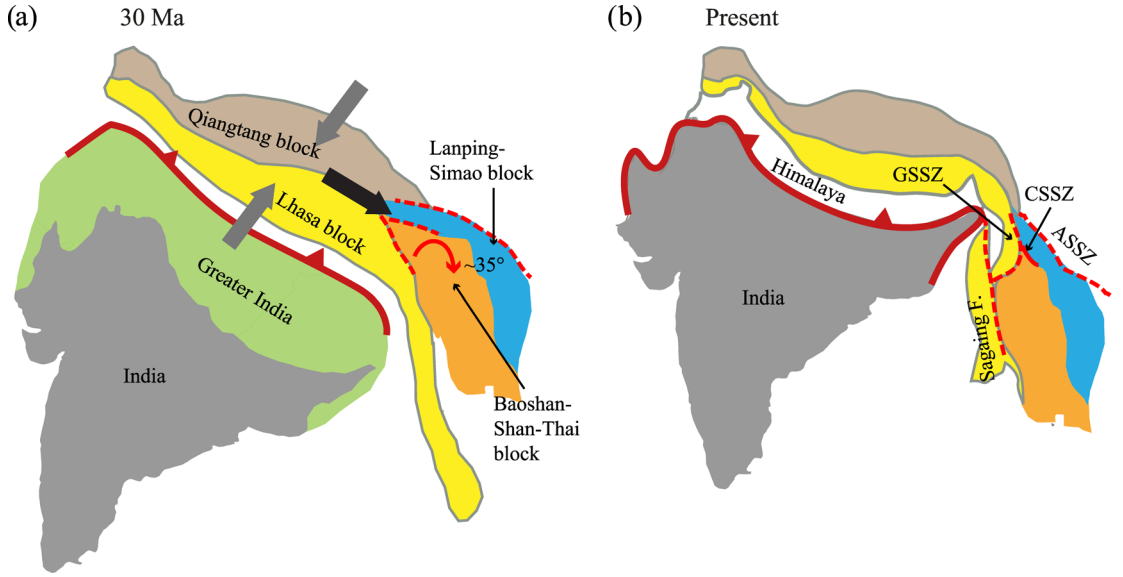


Figure 9. Schematic tectonic evolution of the Baoshan Block since ~30 Ma (modified after Royden *et al.*, 2008). (a) Extrusion of the coupled Baoshan and Lanping-Simao blocks contemporaneous with clockwise rotation due to the northward indentation of India into Eurasia. Onset of shearing along the Chong Shan and Gaoligong Shan shear zones decoupled both blocks; the Baoshan Block continued to rotate and extrude to the SSE leading to the present-day tectonic situation in (b). Shear zones are marked by dashed red lines.

would imply a southward extent of 890 ± 755 km of the Baoshan Block since ~30 Ma. Cogné *et al.* (2013) report a total amount of post-collisional intracontinental convergence between India and Siberia of 2080 ± 630 km since ~50 Ma, consumed by Greater India and Central Asia. Liebke *et al.* (2010) report shortening of the Lhasa Block of 1847 ± 763 km since ~50 Ma. In consideration of the reconstructed palaeopositions of the terranes (e.g. Li *et al.* 2004) and the fact that shortening of the Baoshan Block occurred at Eocene–Oligocene times (Wang *et al.* 2008), 890 km of crustal shortening within the Baoshan Block since ~30 Ma or a more southward position is unlikely but within the statistical uncertainties (± 755 km) possible. This would imply a northward translation, which is disproven by several palaeomagnetic studies, which indicate a southward translation of the Shan-Thai and Lanping-Simao blocks (e.g. Chen *et al.* 1995; Sato *et al.* 2007; Tanaka *et al.* 2008). Therefore, we suggest that the APWP of Besse & Courtillot (2002) is more suitable for our data.

Tapponnier *et al.* (1982) proposed the southeastward tectonic escape of a single rigid lithospheric block between two conjugate strike-slip fault systems, the dextral GSSZ and Sagaing fault zones and the sinistral ASSZ. The crustal flow models imply that large-scale clockwise rotation and crustal thickening accompanied southeastward flow of lower crust (e.g. Royden *et al.* 1997). In these models, the upper crust decoupled from the upper mantle along a weak lower crust, and internal deformation, translation and large-scale differential rotations of smaller crustal blocks accompanied flow of a not-rigid crust (e.g. Wang & Burchfiel 1997). The sinistral ASSZ constituted the eastern boundary of the escaping blocks; it was active during the Oligocene–Miocene and possibly accommodated ~700 km escape of Indochina (e.g. Leloup *et al.* 1995; Chung *et al.* 1997). Low-temperature cooling ages from the Eastern Tibetan Plateau (e.g. Kirby *et al.* 2002; Godard *et al.* 2009) were used to infer the onset of 'crustal flow' at 17–15 Ma. The currently active neotectonic deformation initiated at ~5 Ma (e.g. Lee *et al.* 2003; Taylor *et al.* 2003; Dewane *et al.* 2006; Mahéo *et al.* 2007; Sanchez *et al.* 2010; Ratschbacher *et al.* 2011; Styron *et al.* 2013). Since the onset of neotectonic deformation, the ASSZ has been overprinted by dextral strike-slip faulting (e.g. Jolivet *et al.* 2001; Burchfiel & Wang 2003; Socquet & Pubellier 2005). The dextral GSSZ was likely active between ~20 and 11 Ma; thereafter, dextral shearing was accommodated along the Sagaing Fault (e.g. Socquet & Pubellier 2005; Akciz *et al.* 2008; Lin *et al.* 2009). The sinistral CSSZ was active since at least ~29–24 Ma, terminating at around ~17 Ma (Akciz *et al.* 2008; Wang *et al.* 2006; Zhang *et al.* 2010). New (U–Th)/He and fission-track thermochronology (Sonntag *et al.* 2013) indicates that deformation along the GSSZ and CSSZ continued to 6–3 Ma. The sinistral CSSZ contains evidence for a dextral overprint, indicating shear transfer from the GSSZ (Akciz *et al.* 2008).

Given this time frame, shearing along the CSSZ was contemporaneous with those along the GSSZ and the observed clockwise rotation is best explained by escape of the coupled Baoshan and Lanping-Simao blocks along the ASSZ (Fig. 9a). With the onset of shearing along the CSSZ and GSSZ, the Baoshan Block was decoupled from the Lanping-Simao Block and escaped southeastward between the CSSZ and the GSSZ; the Lanping-Simao Block further escaped between the ASSZ and CSSZ (Akciz *et al.* 2008; Zhang *et al.* 2010; Fig. 9). Thus, the first stage of displacement of the Baoshan Block can be interpreted as rigid block translation and rotation, akin the translation and rotation implied by the escape model (Tapponnier *et al.* 1982), in which the Baoshan Block moved under compression by the northward indentation of India. In a later stage—after 20 Ma—SSE-ward translation and rotation may be related to crustal flow where upper crustal motions are decoupled from those of the lower crust and mantle (e.g. Royden *et al.* 1997; Clark & Royden 2000).

7 CONCLUSIONS

(1) The ^{40}Ar–^{39}Ar spectra of two Nujiang meta-basalts are complex; they indicate that these rocks were either emplaced or metamorphosed at ~30 Ma. The main magnetic carrier—magnetite—is likely of metamorphic origin.

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(2) Two components of magnetization were discriminated in these meta-basalt layers. The LCC likely represents a more recent overprint; it is carried by magnetite. Ti-rich titanomagnetite and magnetite carry the HCC; it represents a syn-folding remanence acquired at ∼30 Ma.

(3) A clockwise rotation of 35.1° ± 12.7° for the Baoshan Block is determined using the APWP of Besse & Courtillot (2002). This corresponds to a rotation rate of 1.17 ± 0.42° Myr^{-1} since remanence acquisition at ∼30 Ma and a palaeolatitude of 28.2° ± 6.5°N. The clockwise rotation of the Baoshan Block is explained by the rotational crustal movements around the EHS during tectonic escape of lithospheric blocks; the driving Indian indenter push northward and the coupled Baoshan and Lanping-Simao blocks escaped along the ASSZ. In a later stage, with the onset of shearing along the CSSZ and the GSSZ, the Baoshan Block decoupled from the Lanping-Simao Block, and both blocks escaped separately. After 20 Ma, eastward crustal flow of Tibetan crust may have contributed to the accumulated rotation.

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APPENDIX: ^{40}AR–^{39}AR GEOCHRONOLOGY

^{40}Ar–^{39}Ar geochronology was conducted at the Argon laboratory (ALF) at the TU Bergakademie Freiberg, Germany. The whole-rock samples were broken to <1-mm fragments and handpicked under a binocular microscope to select fresh material. The fragments were then ultrasonicated in alcohol and repeatedly cleaned in de-ionized water to remove the clay fraction, dried and subsequently wrapped into Al foil. These sample packets were loaded in 5 × 5 mm wells on ∼30-mm Al-discs for irradiation, which was done without Cd shielding for 2.5 hr at the LVR-15 research reactor in Rez, Czech Republic, at a thermal neutron fluence of $\sim 4.8 \times 10^{13}$ n cm^{-2} s^{-1} and a thermal to fast neutron ratio of ∼2.1. Irradiated samples were unwrapped and loaded into small chutes of an autosampler system that allows sample transfer to the high-temperature cell (HTC) without wrapping. Step heating was performed using a Createc® high-temperature cell as furnace; temperature control was managed by an Eurotherm® 3504 controller. Ramp time was 3 min per step at a heating rate of 100 °C min^{-1}, heating time per step was 7 min. Gas purification was achieved within 10 min using two AP10N getter pumps, one at room temperature and one at 400 °C. Ar-isotope compositions were measured in static mode using a GV Instruments ARGUS noble gas mass spectrometer equipped with five Faraday cups and 10^{12} ohm resistors on mass positions 36–39 and a 10^{11} ohm resistor on mass position 40. Typical blank levels are $<2.5 \times 10^{-16}$ mol ^{40}Ar and $<8.1 \times 10^{-18}$ mol ^{36}Ar. Measurement time was 7.5 min per step acquiring 45 scans at 10 s integration time each. Mass bias was corrected assuming linear mass-dependent fractionation and using an atmospheric ^{40}Ar–^{36}Ar ratio of 295.5. For data handling and raw data reduction, an in-house developed MATLAB® software package associated with a MySQL database system was used; isochron, inverse isochron and plateau ages were calculated using ISOPLOT (Ludwig 2008). All ages were calculated using Fish Canyon sanidine as a flux monitor (28.305 ± 0.036 Ma; Renne *et al.* 2010). Errors on ages are 1σ. Decay constants used are those given in Renne *et al.* (2010). Corrections for interfering Ar isotopes were done using $(^{36}Ar/^{37}Ar)_{Ca} = 0.000245$, $(^{39}Ar/^{37}Ar)_{Ca} = 0.000932$, $(^{38}Ar/^{39}Ar)_{K} = 0.01211$, $(^{40}Ar/^{39}Ar)_{K} = 0.00183$ and applying 5 per cent uncertainty.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Table S1. Analytical Details of $^{40}Ar/^{39}Ar$ Geochronology (http://gji.oxfordjournals.org//lookup/suppl/doi:10.1093/gji/ggu009/-/DC1)

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